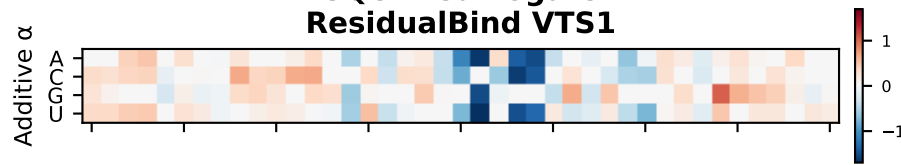
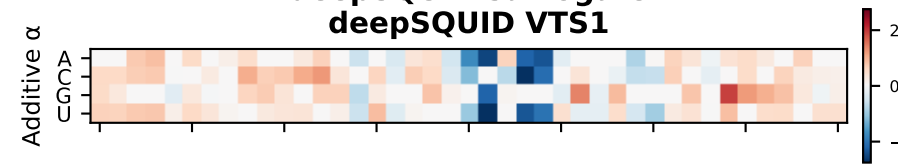


VTS1-high additive and pairwise maps — 10% mutation / 20K fits vs exhaustive saturation

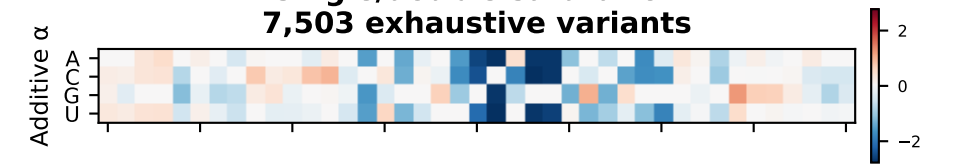
**SQUID surrogate
ResidualBind VTS1**



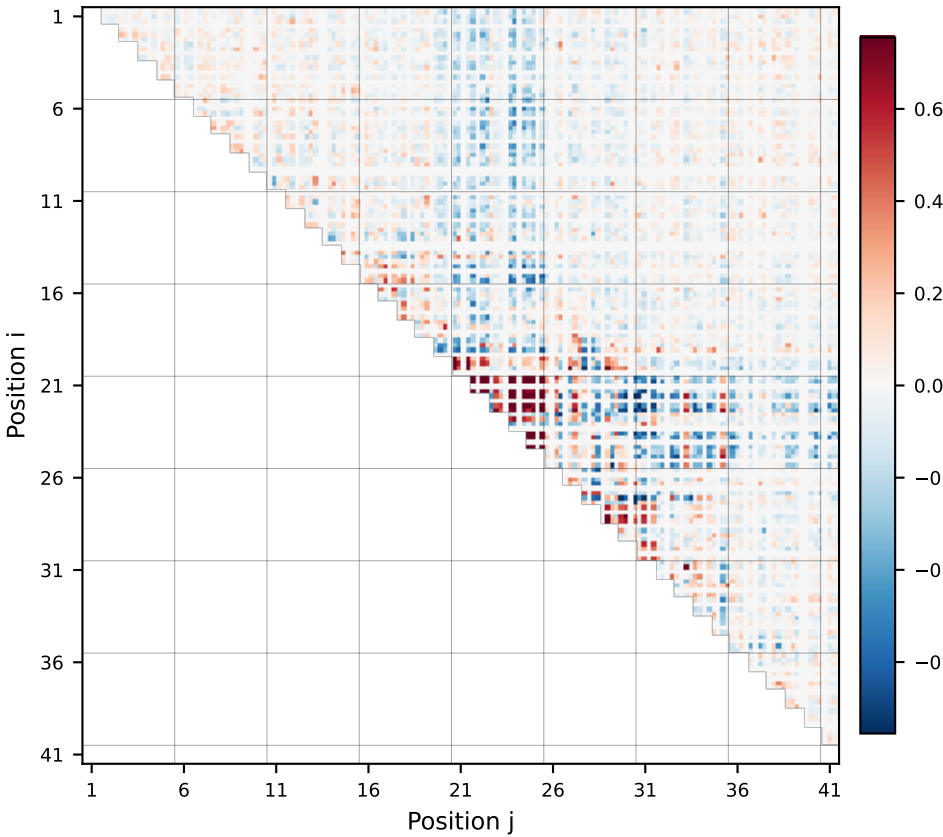
**deepSQUID surrogate
deepSQUID VTS1**



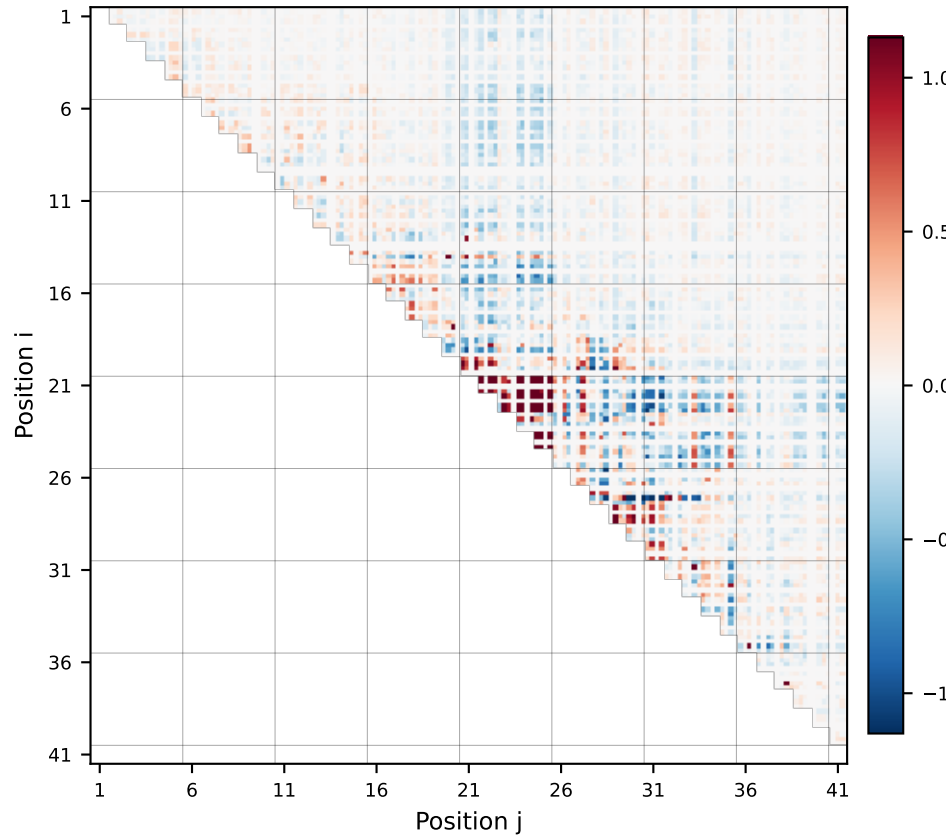
**Single/double saturation
7,503 exhaustive variants**



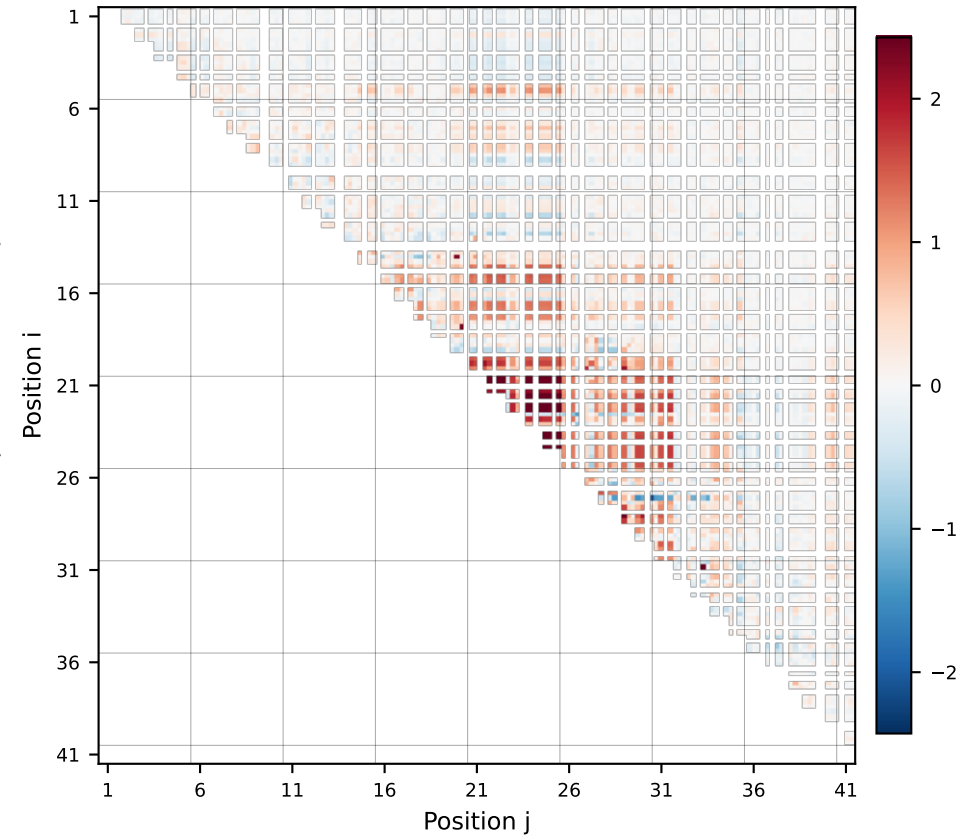
Pairwise matrix



Latent pairwise coefficient j



Latent pairwise coefficient j



Activity epistasis

A/C/G/U states are nested within each position; colors are scaled independently per panel (99.5th percentile).